VPS BASEKIT (VBK) GENERATION 3
DATA SHEET

Figure 1: Gen3 VBK (MAC & SCIP)

The **Vehicle Protection Systems (VPS) Base Kit (VBK)** is designed to provide a Modular Open Systems Architecture (MOSA) within the Survivability domain for U.S. Army ground vehicle platforms. It includes the **Modular Active Controller (MAC)** and the **Safety Control Interface Panel (SCIP)**, enabling modularity, scalability, and reconfigurability for layered capabilities. The VBK supports rapid development for specific missions, platforms, and threats, reducing costs and increasing soldier survivability.

MODULAR ACTIVE CONTROLLER (MAC)

Provides control and coordination for the Vehicle Protection System (VPS).

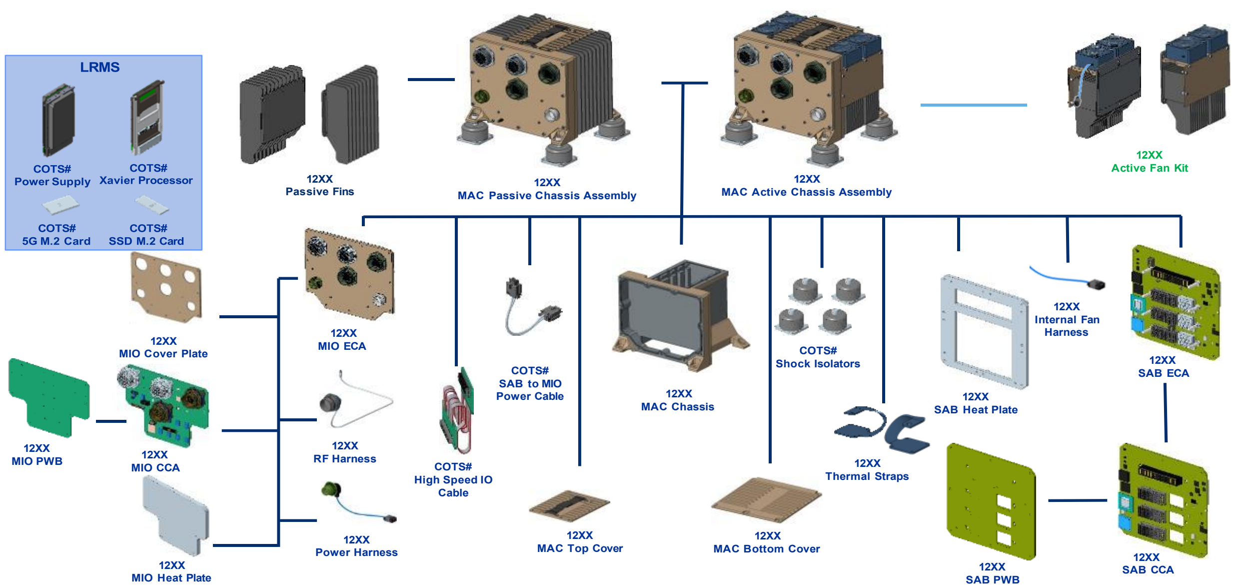

Figure 2: MAC Product Tree

FEATURES:

Open, modular, scalable, and reconfigurable design

Safety-certifiable hardware and software

Interfaces with sensors, countermeasures, and other subsystems

Primary processor – NVIDIA Orin 3U VPX

Supports deterministic Ethernet protocols (TSN)

SPECIFICATIONS:

Data Throughput: 49.9 Gbps (Threshold), 63.9 Gbps (Objective)

Weight: 35.2 lbs

Power Draw:
Steady State: 350W @ 28VDC

Peak: 650W @ 28VDC

Power quality: MIL-STD 1275E

Dimensions:
Height: 8.34 inches

Width: 10.30 inches

Depth: 8.60 inches

Interfaces:

(12) TSN Capable 1-Gigabit Ethernet Interface	(12) TSN Capable 10-Gigabit Ethernet Interface
(1) Serial Debug Interface	(2) Direct Connect Ethernet Interface
(4) RS485 Interface	(4) 3G-SDI Video Inputs
(6) Bidirectional RS422 Interface	(1) HDMI Video Output
(8) Low Speed Opto-Isolated I/O	(1) DisplayPort Video Output
(14) High Speed Opto-Isolated I/O	(1) HD-SDI Video Output
(2) CANbus ISO Interface	(1) Analog RS170 Video Output
(1) GPS Antenna Interface	(1) MIL-STD-1553 Interface

SAFETY CONTROL INTERFACE PANEL (SCIP)

Provides vehicle crew members with control interface, system-safety assurance and status capabilities for the VPS.

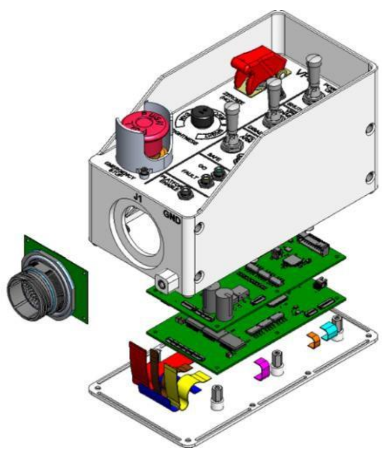

Figure 3: SCIP Exploded View

FEATURES:

- Emergency stop, power control, arming, and zeroization functions
- Power inhibit of safety-critical devices
- LED indicators for system status (e.g., power, fault, armed, safe)
- Rotary dimming control for LED brightness

SPECIFICATIONS:

Weight: 5.20 lbs

Power Draw:
Steady State: 50W @ 28VDC

Peak: 100W @ 28VDC

Dimensions:
Height: 8.351 inches

Width: 4.000 inches

Depth: 4.125 inches

Interfaces:
MIL-DTL-38999 connectors for electrical signals

Discrete signals for system control and feedback

CAN Bus (2 channels), RS232 and Ethernet for data exchange

Mounting provisions for integration into vehicle platform

SYSTEM SPECIFICATION

OPERATIONAL STATES & MODES

Off Mode: Default state when vehicle power is not supplied; all output discretes are inactive

On Mode: System is operational; includes sub-modes such as initialization, standby, active enabled, and fault.

SAFETY FEATURES

Emergency Stop: Terminates power to all subsystems

Zeroization: Clears all volatile and non-volatile data storage

Interlocks: Prevent accidental initiation of hazardous actions

ENVIRONMENTAL SPECIFICATIONS

Operating Temperature: -40°F to 160°F

Storage Temperature: -60°F to 160°F

Altitude:
Operating: 1300 feet below and 15,000 feet above sea level

Non-Operating: Up to 50,000 feet above sea level

Shock and Vibration:
Ballistic Shock: Continuous operation during exposure to shock impulses
Functional Shock: Exposure to 15 ± 3 g, 75.0ms half sine wave

Electromagnetic Compatibility:
MIL-STD-461F and MIL-STD-464C compliance

SYSTEM INTERFACES

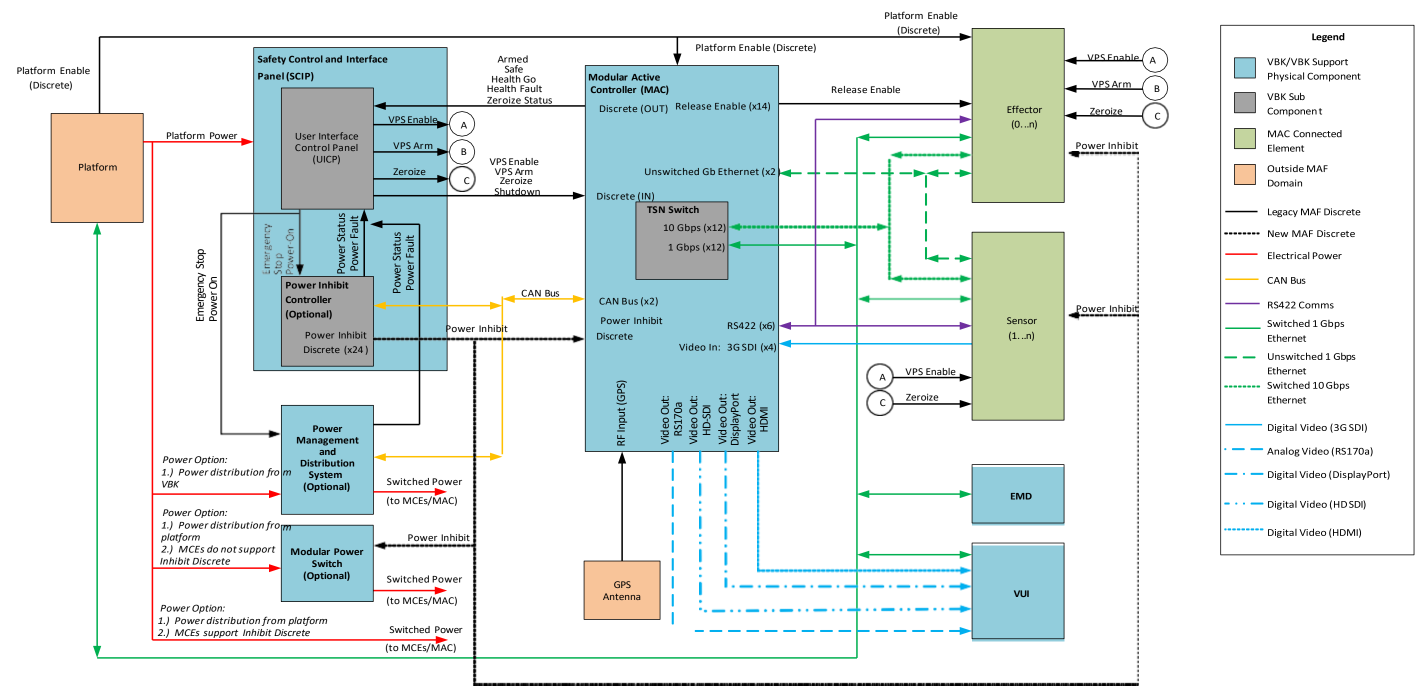

Figure 4: System Interconnect Diagram

RELIABILITY, AVAILABILITY, MAINTAINABILITY

Mean Time Between Failures (MTBF): ≥ 3,500 hours

Mean Time to Repair (MTTR): ≤ 0.5 hours

Maintenance Ratio:

Field Level: ≤ 0.005 maintenance man-hours per operating hour

Sustainment Level: ≤ 0.001 maintenance man-hours per operating hour